

BSE4104 Emerging Trends in Software Engineering

Week 1 Progress Report: Banking Customer Support AI Agent

GROUP B–EVENING

Makerere University

College of Computing and Information Sciences

Week Ending: 4 September 2026

1. Group and Project Overview

GROUP B–EVENING is developing a bounded AI-native customer-support agent for Centenary Bank. The system is intended to help customers obtain reliable information about selected banking products, services, procedures, and support processes. It will also support low-risk activities such as submitting a simulated support request, checking the status of an authorized test case, and escalating sensitive or unresolved matters to a human bank support officer.

The project is designed as a support and information system rather than an autonomous banking decision-maker. The agent will not transfer money, make payments, approve or reject loans, change passwords or PINs, access confidential production databases, or make personalized financial decisions.

2. Work Completed Against Week 1 Objectives

During Week 1, the group completed the initial problem-framing and requirements activities required for the capstone project. A project charter was prepared to define the background, problem statement, target users, system scope, assumptions, constraints, and expected value of AI.

The primary users identified are Centenary Bank customers seeking information or support and bank customer-support officers who handle escalated cases. The main workflow is:

Customer request → intent understanding → knowledge retrieval → next-step decision → approved tool use → response or human escalation.

The group also prepared ten user stories with acceptance criteria. These cover request interpretation, tool execution, session context, persistent memory, failure recovery, traceable outputs, multi-step workflows, workspace operation, user feedback, and execution evidence.

An AI Boundary Matrix was created to distinguish between actions that may be performed by the AI, actions that must remain under deterministic software control, and actions requiring human handling. The repository also contains an initial agent architecture diagram and screenshots showing the project workspace and task-management evidence.

3. Key Engineering Decisions

The group made the following decisions during the first week:

- **Use a controlled knowledge base:** The agent will answer banking questions using approved and traceable banking information rather than relying only on the language model's general knowledge.
- **Use simulated data:** Because the project does not have access to Centenary Bank's confidential systems, customer records, transactions, and support cases will be simulated for academic testing.
- **Keep consequential actions outside AI authority:** Authentication, authorization, validation, financial decisions, credential changes, and transaction processing will remain the responsibility of deterministic software or authorized human staff.
- **Limit the available tools:** The proposed tools are approved knowledge retrieval, support-ticket creation, support-ticket status checking, and escalation to a human officer.
- **Require escalation:** Requests involving fraud, unrecognized transactions, payments, account changes, passwords, PINs, confidential information, or low-confidence answers will be refused or escalated.

These decisions ensure that AI is used for language understanding, retrieval, classification, explanation, and workflow coordination while the system maintains appropriate safety and accountability boundaries.

4. Challenges and Current Response

The main challenge is designing a useful help-desk experience without giving the AI inappropriate authority over banking operations. This has been addressed by defining explicit prohibited actions and escalation rules in the AI Boundary Matrix.

A second challenge is the absence of real Centenary Bank production data. The current response is to use public or team-created banking information and synthetic customer-support records. This allows the workflow to be tested without exposing personal, financial, or confidential information.

A further challenge is ensuring that the agent does not invent banking requirements, charges, or procedures. The planned response is to implement retrieval from an approved corpus, display or record sources, validate tool inputs, and require the agent to acknowledge uncertainty when reliable evidence is unavailable.

5. Repository and Project Evidence

The following evidence has been prepared in the project repository:

- Project charter: docs/requirements/Project_Charter_Final_3_Pages.docx
- User stories and acceptance criteria: docs/requirements/User-Stories-and-Acceptance-Criteria.docx
- AI Boundary Matrix: docs/requirements/AI_Boundary_Matrix.docx
- Initial architecture diagram: docs/architecture/agentarchitecture.png
- GitHub and ClickUp evidence: evidence/screenshots/
- Project structure for prompts, knowledge, source code, tests, traces, and evaluation: prompts/, knowledge/, src/, tests/, and evidence/

The repository is organized to support progressive development from a foundation-model baseline to retrieval-augmented generation, tool use, bounded agency, memory, evaluation, observability, and guardrails.

6. Individual Contribution Summary

Individual ownership is being recorded in the group's ClickUp workspace. The current contribution allocation is:

Member	Task Owned / Evidence
<i>Walusimbi Ashraf</i>	Project charter and problem definition
<i>Mariam Wambui</i>	User stories and acceptance criteria
ClickUp coordination and evidence documentation	
<i>Ssenoga Herman</i>	AI Boundary Matrix and safety constraints
<i>Mugoya Yusuf</i>	Initial architecture and Report

7. Plan for Week 2

In Week 2, the group will implement the smallest useful model-backed capability. The planned activities are to select an accessible language model, document the selection rationale, create and version the initial prompts, and test the baseline using at least ten representative scenarios.

The group will also define expected response formats, source-grounding requirements, refusal behaviour, and failure handling. The results will be used as a baseline before retrieval, tools, and the bounded agent workflow are added in later weeks.

Conclusion

Week 1 established a feasible and safety-conscious foundation for the Centenary Bank customer-support project. The group has defined the target users, core workflow, initial requirements, AI boundaries, repository structure, and evidence locations. The project will continue with a deliberately small and testable scope, using AI where it adds value while retaining deterministic controls and human approval for sensitive banking activities.